

Generation Interconnection Request System Impact Study Report

For Queue Project Q18-1075

Hickory 138 kV
November 28, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1075. The Interconnection Customer proposes a 302.3 MW wind facility interconnecting at the Hickory 138 kV substation in the Virginia area. The request was received on October 09, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 414 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$2,130,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
HICKORY to HARBOR BEACH 138 KV	MIRA VISTA to YARNEN 345 KV L	111.9	360	54.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
HICKORY to HARBOR BEACH 138 KV	Rebuild line with higher-capacity conductor	\$5,362,000	40% (\$2,130,000)
	Total allocated to project		\$2,130,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.